

Phylogeny-Guided Selection of Microbiome Features using Bayesian Prevalence-Aware Multinomial Logit Normal Regression

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SUMMARY: High-throughput microbiome studies increasingly seek to identify microbial taxa associated with host phenotypes. Such analyses are complicated by finite sequencing depth, sparse and heterogeneously prevalent features, and evolutionary relationships that can make nearby taxa respond similarly. Most differential-abundance procedures address only a subset of these structures or analyze taxa independently. We propose a Bayesian prevalence-aware multinomial logit-normal regression model with phylogeny-guided global-local shrinkage (PGL) for operational taxonomic unit selection. The multinomial likelihood represents compositional sampling, while observed presence indicators distinguish the positive-abundance regression from the prevalence pattern. A structured shrinkage prior uses the phylogenetic tree to share information across related effects while retaining taxon-specific estimates and allowing isolated signals to escape global shrinkage. Posterior inference is performed using a Metropolis-within-Gibbs sampler. In simulations varying prevalence, signal magnitude, and phylogenetic alignment, phylogenetic borrowing produced progressively larger power gains as signals became more aligned with the tree, with broadly comparable effect-estimation error and only modest increases in false-positive rate. When signals were phylogenetically unstructured, performance was similar to independent shrinkage. In an inflammatory bowel disease analysis of 183 subjects and 314 operational taxonomic units, the two prior structures yielded strongly concordant posterior means, while PGL identified additional associations concentrated in local phylogenetic neighborhoods. These results illustrate how evolutionary information can improve sensitivity for structured microbiome associations without replacing feature-specific inference by taxonomic aggregation.

KEY WORDS: Microbiome; Differential Abundance Analysis; Bayesian Hierarchical Model; Phylogeny-Guided Global-Local Shrinkage.

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1. Introduction

The human microbiome is associated with diverse aspects of host physiology and disease, motivating the identification of microbial features related to host phenotypes (1; 13; 15). In 16S rRNA gene sequencing studies, denoised reads are commonly organized as operational taxonomic units (OTUs), yielding a sample-by-OTU count table together with a phylogenetic tree inferred from sequence similarity (4; 5). We use OTU as the general statistical term for a sequence-defined microbial feature, including features resolved by modern denoising pipelines. Retaining fine-resolution OTUs avoids obscuring heterogeneous associations among sequence variants assigned to the same named taxon, while taxonomic annotations can subsequently support interpretation at coarser levels. The resulting feature-selection problem is therefore to identify phenotype-associated OTUs while respecting the sampling and evolutionary structure of microbiome data.

This problem has three salient features. First, sequencing depth is finite, so counts within a sample are constrained by their total and are inherently compositional (2; 25). Second, OTU tables are sparse and exhibit heterogeneous prevalence: a zero may reflect biological absence, insufficient sequencing depth, or technical variation, whereas positive counts describe abundance conditional on detection (7; 30). Treating these two processes identically can blur the interpretation of covariate effects. Third, OTUs are not exchangeable biological units. Phylogenetically related microorganisms may share functional traits or respond similarly to host conditions, leading association signals to occur in evolutionarily structured groups (11; 24). A useful feature-selection model should accommodate all three features without requiring prior aggregation of the OTUs.

Existing differential-abundance methods address important subsets of these challenges. Methods based on normalized counts or feature-wise regressions, including DESeq2, edgeR, limma-voom, LefSe, and MaAsLin 2, provide widely used tools for detecting associations un-

der different sampling and modeling assumptions (16; 20; 23; 34; 35). Composition-aware procedures include ALDEx2 and the ANCOM-BC family (10; 17). In particular, ANCOM-BC2 extends bias-corrected differential-abundance analysis to multigroup comparisons, covariate adjustment, and repeated measures (18). More recently, MaAsLin 3 introduced parallel models for feature prevalence and nonzero abundance and can address compositionality using experimental absolute-abundance information or a computational correction (27). These advances offer increasingly flexible feature-wise inference, but they do not use evolutionary relatedness to guide joint shrinkage and selection across OTUs. Phylogenetic information has been incorporated into other microbiome analyses, including community comparison and microbial network estimation (8; 21; 22), but its use in regression-based feature selection remains limited. In particular, there remains a need for an OTU-level framework that embeds prevalence-aware abundance regression within a joint multinomial model and shares information across regression effects according to a given phylogeny.

We propose a Bayesian prevalence-aware multinomial logit-normal regression model to address this need. The observed OTU counts follow a multinomial distribution conditional on the sample sequencing depth, directly representing their compositional dependence. The multinomial probabilities are constructed by normalizing latent OTU-specific pseudo-counts. Each pseudo-count is the product of its observed presence indicator and a positive latent quantity, whose logarithm is related to host covariates through a Gaussian regression. Consequently, zeros do not enter the abundance regression as arbitrarily small positive values, and regression coefficients describe covariate associations with relative abundance among the taxa observed in a sample. This construction retains OTU-level resolution while separating observed prevalence patterns from variation in positive abundance.

To guide estimation and selection, we assign the exposure effects a phylogeny-guided global–local shrinkage prior. A global scale controls overall sparsity for each exposure, local

scales preserve large feature-specific effects, and a phylogenetic correlation matrix couples the standardized effects of related OTUs. The prior can therefore borrow information across the tree without forcing all members of a lineage to have identical effects. Posterior inference simultaneously provides OTU-specific effect estimates and measures of feature importance. We develop a Metropolis-within-Gibbs algorithm for computation and evaluate the proposed approach through simulation studies and microbiome applications. Comparisons with an otherwise identical model that removes phylogenetic correlation isolate the contribution of phylogeny-guided shrinkage.

The remainder of the paper is organized as follows. Section 2 introduces the prevalence-aware multinomial logit-normal model, the structured shrinkage prior, and posterior computation. Section 3 evaluates estimation and feature selection under varying prevalence, signal strength, and phylogenetic alignment. Section 4 presents the microbiome applications, and Section 5 concludes with practical considerations, limitations, and directions for future work.

2. Model

2.1 Prevalence-Aware Multinomial Logit-Normal Regression

For sample $i = 1, \dots, n$, let $\mathbf{y}_i = (y_{i1}, \dots, y_{iK})^\top$ denote the observed counts of K operational taxonomic units (OTUs), and let $N_i = \sum_{k=1}^K y_{ik}$ be the sequencing depth. We model

$$\mathbf{y}_i \mid N_i, \boldsymbol{\pi}_i \sim \text{Multinomial}(N_i, \boldsymbol{\pi}_i), \quad \sum_{k=1}^K \pi_{ik} = 1. \quad (1)$$

This likelihood directly represents the finite-sum constraint on the multivariate counts.

Define the observed presence indicator $\eta_{ik} = I(y_{ik} > 0)$ and a positive latent pseudo-count θ_{ik} . The multinomial probabilities are

$$\pi_{ik} = \frac{\eta_{ik} \theta_{ik}}{\sum_{h=1}^K \eta_{ih} \theta_{ih}}. \quad (2)$$

Thus a taxon not observed in sample i receives zero compositional probability, whereas

positive relative abundances are modeled through θ_{ik} . The indicators η_{ik} are treated as fixed: the proposed model is prevalence-aware but does not specify a probability model for presence or absence. A joint phylogenetic model for prevalence and abundance is discussed as a direction for future work.

Let $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^\top$ contain exposures of scientific interest, and let $\mathbf{z}_i = (z_{i1}, \dots, z_{id})^\top$ contain nuisance covariates, including an intercept when appropriate. We assume

$$\log \theta_{ik} = \mu_{ik} + \epsilon_{ik}, \quad \epsilon_{ik} \stackrel{\text{ind}}{\sim} N(0, \sigma_i^2), \quad (3)$$

where

$$\mu_{ik} = \mathbf{x}_i^\top \boldsymbol{\beta}_{k\cdot} + \mathbf{z}_i^\top \boldsymbol{\gamma}_{k\cdot}. \quad (4)$$

Here β_{kj} is the relative-abundance effect of exposure j for taxon k , and $\gamma_{k\ell}$ is the corresponding nuisance effect. Equations (1)–(4) define a presence-gated multinomial logit-normal regression, related to the logistic-normal construction for compositional data (3). The likelihood is invariant to a common additive shift of the active $\log \theta_{ik}$ within a sample. We use proper centered priors to regularize this invariant direction, so inference is prior-identified and the regression effects are interpreted on the relative compositional scale. Details are provided in Web Appendix C.

2.2 Phylogeny-Guided Global–Local Shrinkage

For exposure j , collect its taxon-specific effects in $\boldsymbol{\beta}_{\cdot j} = (\beta_{1j}, \dots, \beta_{Kj})^\top$ and write

$$\boldsymbol{\beta}_{\cdot j} = \tau_j \boldsymbol{\Lambda}_j \tilde{\boldsymbol{\beta}}_{\cdot j}, \quad \boldsymbol{\Lambda}_j = \text{diag}(\lambda_{1j}, \dots, \lambda_{Kj}), \quad \tilde{\boldsymbol{\beta}}_{\cdot j} \sim N_K(\mathbf{0}, \mathbf{C}). \quad (5)$$

Here τ_j is an exposure-specific global scale, λ_{kj} is a taxon-specific local scale, and the phylogenetic correlation matrix $\mathbf{C}$ couples the standardized effects. Conditional on these scales,

$$\text{Cov}(\beta_{k_1j}, \beta_{k_2j} \mid \tau_j, \lambda_{k_1j}, \lambda_{k_2j}) = \tau_j^2 \lambda_{k_1j} \lambda_{k_2j} C_{k_1k_2}. \quad (6)$$

Equation (6) clarifies the distinct roles of the three components. If $C_{k_1 k_2} = 0$, the prior does not directly couple the two effects. If the taxa are phylogenetically correlated, a small global scale jointly suppresses their effects, whereas a large local scale allows a particular effect to escape global shrinkage. When both local scales are appreciable, related signals can be learned jointly through $\mathbf{C}$; when either is strongly shrunk, the induced covariance is attenuated.

This conditional-dependence mechanism also explains why ordinary global–local scales are not sufficient for our purpose. For a correlated signal–noise pair, the global scale must concentrate strongly near zero to prevent the signal from propagating into the null coordinate, while the local scale of the signal must have a sufficiently heavy upper tail to overcome that global shrinkage. When both coordinates are signals, large local scales should preserve their phylogenetically structured co-variation; when both are noise, the scales should rapidly collapse their marginal variances and covariance. Thus dependent global–local shrinkage places simultaneous and particularly strong demands on behavior near zero and in the upper tail (41).

We meet these demands by assigning both the global and local scales positive log-Cauchy priors,

$$\tau_j \stackrel{\text{ind}}{\sim} \text{LC}^+, \quad \lambda_{kj} \stackrel{\text{ind}}{\sim} \text{LC}^+, \quad p_{\text{LC}^+}(s) = \frac{1}{s\{\pi^2 + \log^2(s)\}} I(s > 0). \quad (7)$$

Equivalently, $\log s \sim \text{Cauchy}(0, \pi)$. The density has a pole at the origin and a super-heavy upper tail: the former produces strong shrinkage of small effects, while the latter limits overshrinkage of large signals. These properties, including tail robustness and sparse posterior contraction, distinguish the log-Cauchy construction from standard half-Cauchy global–local priors (40). The phylogenetic dependence therefore motivates, rather than merely accompanies, the choice of log-Cauchy scales.

The same dependence propagates to the abundance layer. Let $c_{ik} = \eta_{ik}\theta_{ik}$, $\mathbf{H}_j = \tau_j \mathbf{\Lambda}_j$,

$\mathbf{\Gamma} = (\gamma_{k\ell})_{1 \leq k \leq K, 1 \leq \ell \leq d}$, $\mathbf{m}_i = \mathbf{\Gamma} \mathbf{z}_i$, and

$$\mathbf{V}_{\mu,i} = \sum_{j=1}^p x_{ij}^2 \mathbf{H}_j \mathbf{C} \mathbf{H}_j.$$

For $k_1 \neq k_2$,

$$\begin{aligned} & \text{Cov}(c_{ik_1}, c_{ik_2} \mid \boldsymbol{\eta}_i, \text{hyper}) \\ &= \eta_{ik_1} \eta_{ik_2} \exp \left\{ m_{ik_1} + m_{ik_2} + \frac{V_{\mu,i}(k_1, k_1) + V_{\mu,i}(k_2, k_2)}{2} + \sigma_i^2 \right\} [\exp\{V_{\mu,i}(k_1, k_2)\} - 1]. \end{aligned} \quad (8)$$

Hence the cross-taxon covariance vanishes when either taxon is absent or when the active exposures induce no cross-taxon coupling. Its magnitude otherwise depends jointly on prevalence, exposure values, abundance, global–local scales, and phylogeny. Web Appendix B gives the derivation and further interpretation.

Finally, we assign independent priors

$$\gamma_{k\ell} \sim N(0, 0.71^2), \quad \sigma_i \sim N^+(0, 0.63^2), \quad (9)$$

whose prior-predictive calibration is given in Web Appendix D.

2.3 Construction of the Phylogenetic Correlation Matrix

Our construction is inspired by the graph-Laplacian representation introduced for phylogeny-aware community comparison in GLaD (42), but is adapted here to define prior dependence among taxon-specific regression effects.

Represent the phylogenetic tree as a connected, undirected graph containing both observed tips and internal ancestral nodes. For its q nodes, define the symmetric weighted adjacency matrix $\mathbf{A}$ by

$$\begin{aligned} A_{uv} &= \begin{cases} \ell_{uv}, & \text{if nodes } u \text{ and } v \text{ are joined by a branch of length } \ell_{uv}, \\ 0, & \text{otherwise,} \end{cases} \\ \mathbf{D} &= \text{diag} \left(\sum_v A_{1v}, \dots, \sum_v A_{qv} \right). \end{aligned} \quad (10)$$

We form the symmetric normalized Laplacian and its Moore–Penrose pseudoinverse,

$$\mathbf{L}_{\text{norm}} = \mathbf{I}_q - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}, \quad \mathbf{B} = \mathbf{L}_{\text{norm}}^+. \quad (11)$$

The normalization accounts for the total branch length incident to each node, and the pseudoinverse propagates local connections through the full ancestral graph. Let $\mathbf{B}_{\text{tip}}$ be the $K \times K$ submatrix for the observed OTUs. With

$$\mathbf{S} = \text{diag}\{B_{\text{tip},11}^{-1/2}, \dots, B_{\text{tip},KK}^{-1/2}\}, \quad \mathbf{C}_0 = \mathbf{S} \mathbf{B}_{\text{tip}} \mathbf{S},$$

the implementation takes $\mathbf{C}$ to be the nearest positive-definite correlation matrix to $\mathbf{C}_0$. Nonfinite or nonpositive branch lengths are first replaced by a small positive floor. Web Appendix A provides the full construction and implementation.

A conventional alternative is the Brownian-motion covariance returned by `vcv.phylo` in the `ape` package (29). Its diagonal contains root-to-tip lengths and each off-diagonal contains the branch length shared by two taxa from the root to their most recent common ancestor; diagonal standardization then gives a correlation matrix. Although this construction uses ancestry, it compresses the topology into pairwise shared-path lengths. Many OTU pairs descending from the same ancestral split therefore receive tied covariance values, and the resulting matrix can be ill-conditioned in large, closely related phylogenies (14). Our construction instead retains the complete node–edge representation before tip extraction, so both branch length and the way a branch participates in the surrounding tree contribute to dependence. Branch-based summaries such as UniFrac remain informative, while the graph representation additionally encodes how branches are coupled through internal nodes and extended evolutionary paths (21; 22).

2.4 Posterior Computation

We use a Metropolis-within-Gibbs sampler that combines scalar log-scale Metropolis updates with Gaussian block updates. Let $u_{ik} = \log \theta_{ik}$ and $S_i = \sum_{h=1}^K \eta_{ih} \exp(u_{ih})$. For

each active coordinate, propose $u_{ik}^* = u_{ik} + \varepsilon$, where $\varepsilon \sim N(0, s_u^2)$, and replace only the corresponding term in S_i . The log acceptance ratio is

$$\begin{aligned} \log r_u = & y_{ik}(u_{ik}^* - u_{ik}) - N_i(\log S_i^* - \log S_i) \\ & - \frac{(u_{ik}^* - \mu_{ik})^2 - (u_{ik} - \mu_{ik})^2}{2\sigma_i^2}. \end{aligned} \quad (12)$$

Thus the multinomial contribution and Gaussian abundance layer are evaluated without recomputing a K -term normalizing sum.

For exposure j , define $\mathbf{H}_j = \tau_j \mathbf{\Lambda}_j$ and let $\mathbf{r}_i^{(j)}$ be the residual after removing all terms except $x_{ij} \mathbf{H}_j \tilde{\boldsymbol{\beta}}_j$. The standardized effects have the block Gaussian full conditional

$$\tilde{\boldsymbol{\beta}}_j \mid \text{rest} \sim N_K(\mathbf{m}_j, \mathbf{V}_j), \quad \mathbf{V}_j^{-1} = \mathbf{C}^{-1} + \sum_{i=1}^n \frac{x_{ij}^2}{\sigma_i^2} \mathbf{H}_j^2, \quad (13)$$

where

$$\mathbf{m}_j = \mathbf{V}_j \sum_{i=1}^n \frac{x_{ij}}{\sigma_i^2} \mathbf{H}_j \mathbf{r}_i^{(j)}. \quad (14)$$

Updating this vector jointly preserves the phylogenetic dependence in $\mathbf{C}$. Each nuisance-effect vector $\boldsymbol{\gamma}_\ell$ is updated analogously from its Gaussian full conditional.

The positive scales are updated on the log scale. For example, $a_j = \log \tau_j$ is proposed from $N(a_j, s_\tau^2)$. Under Equation (7), the prior density for a_j after the change of variables is proportional to

$$\{\pi^2 + a_j^2\}^{-1};$$

the factor τ_j^{-1} in the positive log-Cauchy density cancels the Jacobian. Consequently, the log-Cauchy contribution to the Metropolis ratio is

$$-\log\{\pi^2 + (a_j^*)^2\} + \log\{\pi^2 + a_j^2\}.$$

The local scales λ_{kj} are handled identically, while each σ_i uses a log-scale Metropolis update under its half-normal prior. This parameterization avoids positivity constraints and directly implements the super-heavy-tailed priors motivated in Section 2.2.

Proposal variances are adapted during warm-up toward an acceptance rate of 0.40 and then fixed for retained posterior sampling. The implementation caches S_i , fitted means, and residual components to avoid repeated $O(K)$ and $O(pK)$ calculations inside scalar updates. Web Appendix E gives all full conditional distributions and Metropolis-Hastings acceptance ratios. Dataset-specific iteration counts, burn-in, thinning, and convergence diagnostics are reported with the simulation and real-data analyses.

3. Simulation

3.1 Design and evaluation

We conducted a simulation study to determine when phylogenetic borrowing improves feature selection and whether this improvement compromises estimation of taxon-specific exposure effects. The primary comparison was between the proposed phylogeny-guided global-local model (PGL) and an otherwise identical independent global-local model (PIND). The two models used the same likelihood, gating variables, priors, initialization, MCMC settings, and simulated data; PIND differed only by replacing the phylogenetic correlation matrix $\mathbf{C}$ with the identity matrix. This comparison isolates the contribution of the phylogenetic dependence in the shrinkage prior. We also included DESeq2 as a widely used feature-wise count-regression benchmark (20).

For each replicate, we generated a random phylogenetic tree with $K = 200$ tips and constructed $\mathbf{C}$ from its normalized graph Laplacian as described in Section 2.3. Let $A \in \{0, 0.5, 1\}$ denote the *phylogenetic alignment* of the exposure signal: among the 60 active taxa, a proportion A was placed in nonoverlapping clades of 5–20 tips and the remainder was selected without reference to the tree. Thus $A = 0$ represents an unstructured active set, $A = 0.5$ a mixture of clustered and unstructured signals, and $A = 1$ complete phylogenetic alignment. Effects within an active clade shared a randomly generated center, with correlated

within-clade deviations having standard deviation $0.10\sqrt{V}$, where $V \in \{0.5, 1\}$ is the signal variance. Unstructured active effects were generated independently with variance V . To avoid signals arbitrarily close to zero, the magnitude of each generated clade center or independent effect was bounded below by $0.50\sqrt{V}$. All remaining exposure coefficients were zero, giving 30% signal prevalence.

Presence indicators were generated through a phylogenetically correlated probit construction. Specifically, a K -variate standard normal vector with correlation $\mathbf{C}$ was drawn for each sample and thresholded to attain marginal presence probability $P \in \{0.25, 0.50\}$. This construction generates realistic co-occurrence among related taxa while allowing us to vary the overall sparsity of the observed table. Conditional on the presence indicators, the remaining data-generating mechanism followed the proposed abundance model. We generated one target exposure and five nuisance covariates independently from standard normal distributions. The nuisance coefficients were independent $N(0, 0.3^2)$ variables and the sample-specific residual standard deviations were uniform on $(0.4, 0.6)$. Latent log pseudo-counts were generated as

$$u_{ik} = x_i\beta_k + \mathbf{z}_i^\top \boldsymbol{\gamma}_k + \sigma_i \epsilon_{ik}, \quad \epsilon_{ik} \stackrel{\text{iid}}{\sim} N(0, 1),$$

and the multinomial probabilities were obtained by normalizing $\eta_{ik} \exp(u_{ik})$. Finally, each sample contained 50,000 multinomial reads.

The main experiment used $N = 50$ and the full factorial combination of the three alignment levels, two presence probabilities, and two signal variances, giving 12 settings. We targeted 100 Monte Carlo replicates per setting. A sample-size experiment repeated the setting $P = 0.50, V = 1$ with $N = 100$ and $A \in \{0, 1\}$. Table 1 summarizes the design. Within each paired block, alignment settings shared the tree, presence variables, covariates, nuisance effects, and random-number streams whenever the signal-construction step was successful. Web

Appendix F provides the complete data-generating algorithm and the number of available replicates in each setting.

[Table 1 about here]

For PGL and PIND, we ran 40,000 MCMC iterations, retained every fourth draw, and discarded the first half of the stored draws. Both global and local scales had the log-Cauchy prior in Equation (7); the nuisance and residual-scale priors were those in Equation (9). Proposal scales were adapted during warm-up toward an acceptance rate of 0.40. DESeq2 was fitted to the same count table with the target exposure and five nuisance covariates, using a local dispersion trend; unit pseudo-counts were used to permit size-factor estimation in the sparse tables.

Our primary Bayesian selection rule declared taxon k active when its 95% equal-tailed posterior credible interval for β_k excluded zero. DESeq2 used the Benjamini–Hochberg adjusted p -value threshold 0.05. We evaluated power, false-positive rate (FPR), false discovery rate (FDR), F_1 score, and Matthews correlation coefficient. Estimation was evaluated using the root mean squared error (RMSE) and mean absolute error of the posterior mean, both over all taxa and separately over active and null taxa. Because PGL and PIND were fitted to the same replicate, we additionally report the active-set RMSE ratio $\text{RMSE}_{\text{PGL}}/\text{RMSE}_{\text{PIND}}$ and paired differences in power and FPR. Values of the RMSE ratio below one favor PGL. The posterior-neighborhood rule $pSN < 0.50$, posterior medians, signed recovery, additional metrics, and MCMC diagnostics are reported in Web Appendix F.

3.2 Results

The advantage of phylogenetic borrowing was governed by the alignment between the active effects and the tree (Figure 1). When $A = 0$, PGL and PIND had nearly identical power: the mean paired differences ranged from -0.004 to 0.016 across the four (P, V) combinations. At $A = 0.5$, the power gain ranged from 0.044 to 0.159 , and at $A = 1$ it ranged from 0.121 to

0.368. The largest gains occurred in the low-prevalence settings, where individual taxon-level information was weakest. For example, with $P = 0.25, V = 1$, power at $A = 1$ was 0.691 for PGL and 0.323 for PIND. Thus PGL did not produce a general advantage for arbitrary sparse signals; its gain emerged as the true signal became increasingly compatible with the phylogenetic structure used by the prior.

The increased sensitivity was accompanied by a modest increase in false positives. Relative to PIND, the mean FPR increase for PGL was 0.013–0.037 in the partially or fully aligned settings, while the absolute PGL FPR generally remained between 0.013 and 0.041. DESeq2 had lower power than PGL in most moderate- and high-prevalence aligned settings and substantially larger FDR, often between 0.4 and 0.6. Taken together, the power and FDR panels show that the additional PGL discoveries were not reproduced by simply using a more liberal feature-wise procedure.

[Figure 1 about here]

Effect estimation remained broadly comparable between the two Bayesian models (Figure 2). With unstructured signals, the mean active-set RMSE ratio was 1.04–1.07, indicating a small cost for imposing an unhelpful correlation structure. Under partial alignment, the ratio was close to one and favored PGL in three of four settings. Under complete alignment, PGL improved active-effect RMSE in most settings, with typical mean ratios between 0.88 and 0.98. The paired selection trade-off gives the same message from a complementary perspective: points moved upward as alignment increased, representing substantial power gains, while moving only modestly to the right in FPR. These results support the intended role of the structured prior—borrowing strength for phylogenetically coherent signals while retaining taxon-specific estimation through the local scales.

[Figure 2 about here]

4. Real Data Analysis

We applied PGL to a publicly available 16S rRNA sequencing study of inflammatory bowel disease (IBD). The study combined microbiome measurements with detailed clinical meta-data to characterize ulcerative colitis (UC) and Crohn’s disease (CD), including heterogeneity related to disease location and severity (12; 26). Our objective was not to establish new disease biomarkers, but to illustrate how the structured prior changes OTU-level inference relative to an otherwise identical independent analysis.

We restricted the data to samples labeled UC or CD and retained OTUs observed in at least 5% of these samples. Samples without complete age at diagnosis, sex, or height information were excluded. Tips having essentially zero pairwise phylogenetic distance were merged by summing their counts, after which their redundant tree tips were removed. The resulting analysis included $N = 183$ samples and $K = 314$ OTUs. The target exposure was an indicator for CD, with UC as the reference category. Standardized age at diagnosis, sex, and standardized height were included as nuisance covariates. Thus a negative exposure coefficient denotes greater relative abundance in UC under the adopted coding.

We fitted PGL and PIND to the same processed count table. Both models used the same likelihood, gating indicators, priors, initial state, MCMC seed, and proposal adaptation; PIND differed only by replacing the phylogenetic correlation matrix $\mathbf{C}$ with $\mathbf{I}_K$. Each chain contained 30,000 iterations, every fifth iteration was retained, and the first half of the stored draws was discarded, leaving 3,000 posterior draws. As in the primary simulation analysis, an OTU was selected when its 95% equal-tailed posterior credible interval excluded zero. Additional preprocessing details, complete feature-level results, threshold sensitivity, and MCMC diagnostics are given in Web Appendix G.

The OTU-specific posterior means were broadly concordant between the two models (Pearson correlation 0.853; Figure 3A). PGL selected 130 OTUs and PIND selected 6; 4 were

selected by both models, 126 only by PGL, and 2 only by PIND. Of the 130 PGL selections, 124 had negative posterior means and 6 had positive posterior means. The larger number of PGL selections persisted over posterior credible levels from 80% to 99% (Figure 3B), rather than arising only at the primary 95% cutoff.

[Figure 3 about here]

To visualize local borrowing without choosing a lineage by biological name, we used a prespecified display rule. Among subtrees containing 10–20 tips, we selected the subtree with the largest proportion of 95%-interval PGL-only discoveries, breaking ties by the number of such discoveries and then the mean absolute PGL posterior effect. The selected 10-tip subtree contained 9 PGL-only discoveries. Figure 4 shows all tips in this subtree, including the nonselected tip. Posterior means were generally directionally similar under PGL and PIND, whereas the PGL intervals more often excluded zero across neighboring OTUs. The display illustrates the intended role of the prior: related features can share information while retaining distinct OTU-specific effect estimates.

[Figure 4 about here]

These selections should be interpreted as a demonstration of model-based sensitivity rather than as validated clinical biomarkers. The comparison is cross-sectional, residual clinical heterogeneity may remain, and a small number of feature-specific posterior summaries had low effective sample sizes. We therefore emphasize the global agreement and the local borrowing pattern rather than individual taxon-level claims.

5. Discussion

We developed a prevalence-aware multinomial logit-normal model for OTU-level differential abundance analysis with phylogeny-guided global–local shrinkage. The multinomial likelihood respects the compositional sampling mechanism, while the observed presence indicators prevent structural zeros from being interpreted as positive-abundance observa-

tions. For each exposure, the global scale favors a sparse collection of associations, the local scales allow individual OTUs to escape that shrinkage, and the correlation matrix $\mathbf{C}$ allows related standardized effects to share information without requiring identical coefficients. This combination is important: borrowing is selective rather than equivalent to taxonomic aggregation, so the output remains a set of OTU-specific effect estimates. Moreover, PIND provides a particularly transparent comparator because it changes only $\mathbf{C}$ to $\mathbf{I}_K$; differences between the two fits can therefore be attributed to the structured prior rather than to a different likelihood or zero-handling rule.

The empirical results reflect the intended, conditional advantage of this construction. In simulation, PGL and PIND behaved similarly when the active signals were phylogenetically unstructured. As alignment increased, PGL achieved progressively larger power gains, especially for less prevalent features, with only modest increases in false-positive rate and generally comparable, or under strong alignment improved, effect-estimation error. Thus the tree was useful when it contained relevant information but caused little loss when it did not. The IBD analysis showed the corresponding inferential pattern in observed data: posterior means from PGL and PIND were strongly concordant, yet PGL produced substantially more intervals excluding zero and revealed a local subtree in which neighboring OTUs had similar directions but retained distinct estimates. These findings demonstrate increased model-based sensitivity rather than independent biological validation. The absence of selected features in the secondary mouse analysis (Web Appendix H) is also informative: the structured prior does not mechanically force discoveries in every dataset.

The sampler was practical for the moderate dimensions considered here, but its mixing was not uniform across parameters. The C++ Metropolis-within-Gibbs implementation required approximately 25 and 23 minutes for the 30,000-iteration IBD PGL and PIND fits, respectively, on the workstation used for the analysis. Post-warm-up acceptance rates

for the principal random-walk updates were near the targeted 0.40 in both fits. Median effective sample sizes for the IBD exposure coefficients were about 93 for PGL and 151 for PIND among 3,000 retained draws, whereas the minima were approximately 9 and 12. The latter values are consistent with the few unstable feature-level summaries noted in Section 4; longer runs and multiple independently initialized chains are advisable when inference about particular OTUs is important. Jointly updating each standardized exposure-effect vector helps preserve the dependence structure, but dense K -dimensional Gaussian operations and the sample-by-feature latent abundance updates will eventually limit scaling. Sparse tree-based factorizations, more efficient gradient-based or elliptical updates, and parallel chains are natural computational improvements.

The main modeling limitation is that the method is prevalence-aware rather than a joint model for prevalence and positive abundance. Specifically, η_{ik} is treated as an observed gate, so β_{kj} describes relative abundance conditional on observed presence and does not estimate how an exposure changes the probability that OTU k occurs. A joint hurdle model could add a Bernoulli or probit prevalence layer, but whether and how that layer should borrow along the phylogeny is a substantive question rather than a purely technical choice. There is biological motivation for such borrowing: marine bacterial communities can be phylogenetically clustered under habitat filtering (32), phylogenetic signal in ecological niches has been observed across multiple ecosystems (38), and, within the human microbiome, close relatives tended to co-occur within a body-site niche (9). These results are compatible with conserved traits jointly affecting colonization or persistence. They do not, however, imply a universal monotone relationship between evolutionary distance and co-presence: community assembly is habitat- and scale-dependent, and observed co-occurrence may reflect shared environment, dispersal, competition, facilitation, or detection rather than direct interaction (9; 24).

Using the same phylogenetic latent process for prevalence and abundance would also induce cross-process association through the tree. That coupling may be desirable when conserved traits govern both establishment and growth, but it may be misleading when transmission and niche entry determine occurrence while host conditions determine relative abundance. We therefore regard a joint prevalence–abundance model as an important future direction, but not as an automatic replacement for the present gate. A useful extension would allow separate prevalence and abundance kernels and an estimable cross-process correlation, including zero correlation as a special case. Repeated samples, technical controls, or absolute-abundance measurements would help distinguish biological absence from imperfect detection and would make that dependence more identifiable. Comparing shared-kernel, separate-kernel, and independent prevalence specifications would then test phylogenetic dependence instead of assuming it.

Two additional limitations merit attention. First, $\mathbf{C}$ is treated as known and depends on a single estimated tree and a chosen tree-to-correlation transformation. Tree uncertainty, branch-length error, limited marker-gene resolution, or phylogenetically labile traits can all make the imposed borrowing imperfect. Sensitivity analyses over plausible trees and kernels, or a hierarchical treatment of kernel uncertainty, would make this component more robust. Second, the 95% credible-interval rule was chosen as a transparent summary of posterior sign certainty rather than to target a prespecified global error rate. If explicit error-rate control is desired, the existing posterior draws provide a direct route to calibration. For example, OTUs could be ordered by their local false sign rates, and the largest set whose average posterior false sign probability does not exceed a target level could be selected. The resulting threshold could additionally be assessed, or tuned for frequentist false-discovery-rate behavior, using null or posterior-predictive simulations that reproduce the fitted prevalence, compositional, and phylogenetic structure. This is a feasible extension because the current sampler already

provides the OTU-specific sign probabilities and their joint posterior dependence; calibration would add a decision layer rather than require a new abundance model. External cohorts would then remain important for assessing replicability and clinical generalizability.

In summary, PGL provides a coherent way to use evolutionary information in sparse compositional regression while preserving feature-specific inference and an explicit interpretation conditional on presence. Its value is not that phylogeny must improve every analysis, but that it supplies principled and locally adaptive borrowing when related taxa respond similarly. The simulation and real-data examples show both sides of that claim and position the framework as a useful foundation for richer joint models of microbial occurrence and abundance.

SUPPLEMENTARY MATERIALS

Web Appendices A–G, referenced in Sections 2–4, are available with this paper at the Biometrics website on Wiley Online Library.

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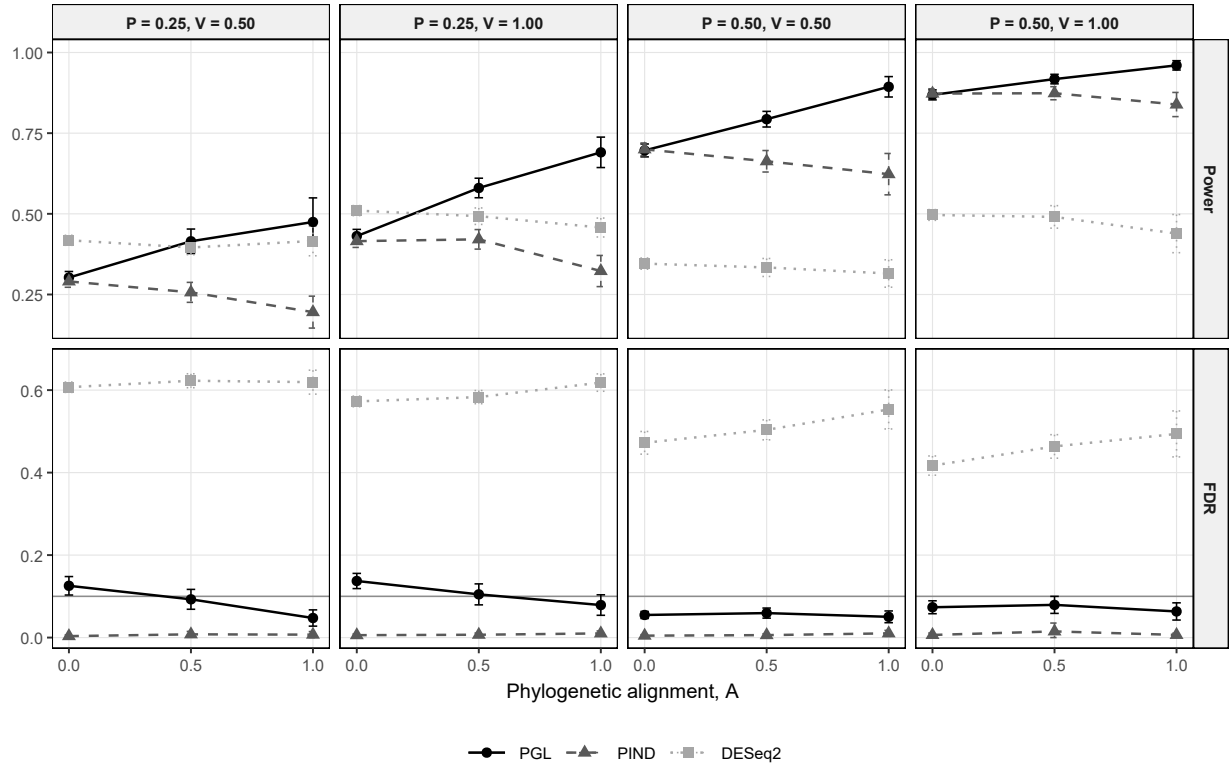


Figure 1. Feature-selection performance in the main simulation experiment. Points show Monte Carlo means and bars show 95% Monte Carlo intervals. PGL and PIND select a feature when its 95% posterior credible interval excludes zero; DESeq2 uses a Benjamini–Hochberg adjusted p -value below 0.05. The horizontal gray line in the FDR panels marks 0.10. A is the proportion of active taxa aligned with phylogenetic clades, P is the marginal presence probability, and V is the signal variance.

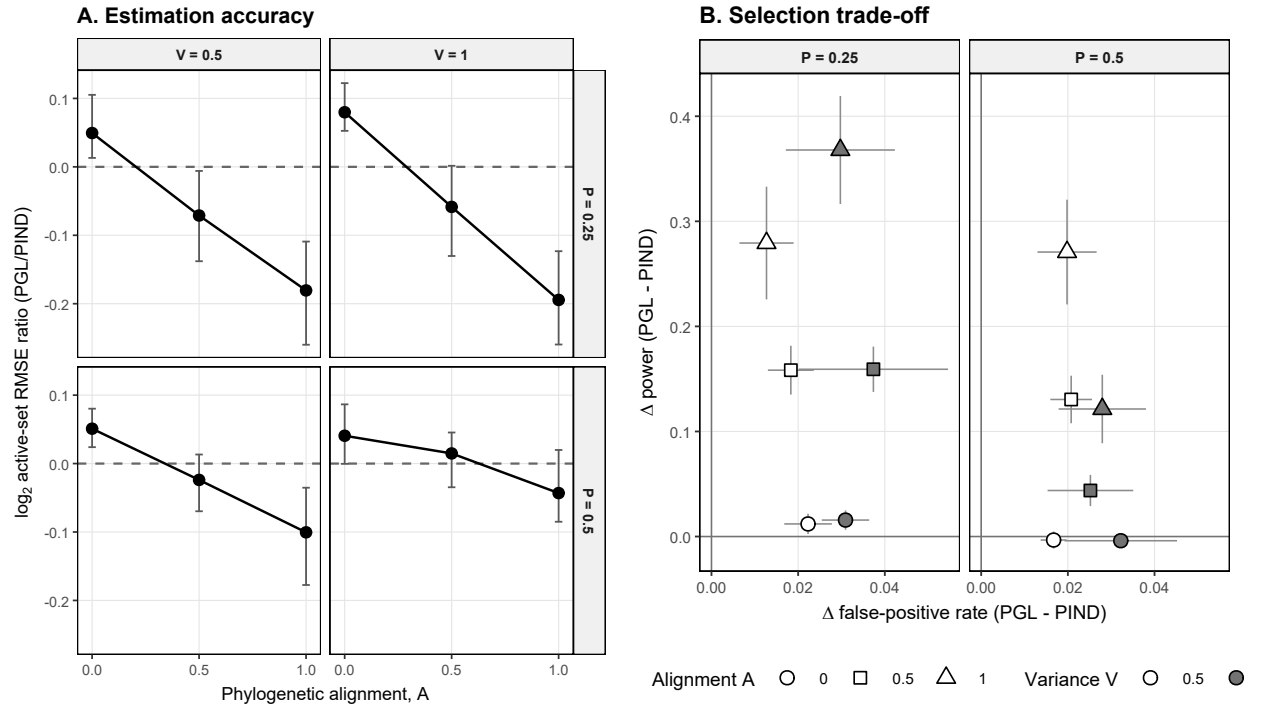


Figure 2. Paired comparison of PGL and PIND in the main simulation experiment. Panel A shows the median and interquartile range, over paired replicates, of the log-base-2 ratio of active-set RMSE; values below zero favor PGL. Panel B shows the mean paired gain in power against the mean paired increase in false-positive rate under 95% credible-interval selection. Horizontal and vertical segments show 95% Monte Carlo intervals for the respective mean differences. A , P , and V denote phylogenetic alignment, marginal presence probability, and signal variance, respectively.

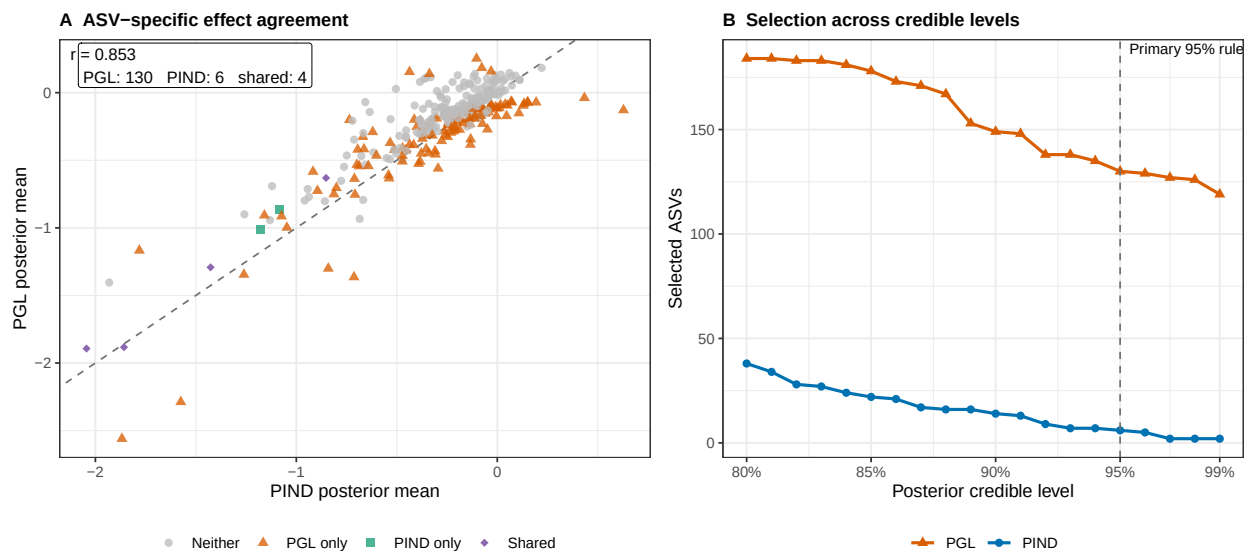


Figure 3. Comparison of phylogeny-guided and independent shrinkage in the IBD application. Panel A compares OTU-specific posterior means under PIND and PGL. Colors and symbols indicate selection by 95% equal-tailed posterior credible intervals. Panel B shows the number of selected OTUs over posterior credible levels from 80% to 99%; the vertical dashed line marks the primary 95% rule. The target coefficient contrasts Crohn's disease with ulcerative colitis, so negative values indicate greater relative abundance in ulcerative colitis under the adopted coding.

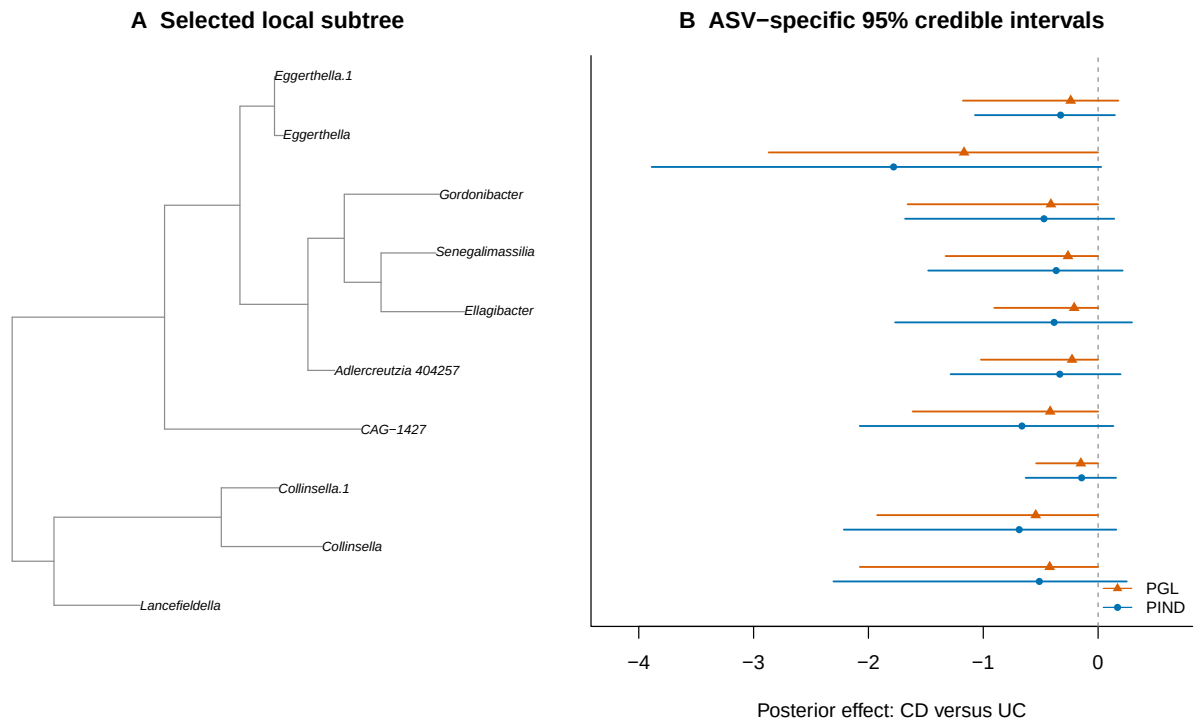


Figure 4. Illustration of local phylogenetic borrowing in the IBD application. Panel A shows the 10-tip subtree chosen by a prespecified rule that maximized the density of PGL-only discoveries among subtrees containing 10–20 tips; all tips in the selected subtree are displayed. Panel B shows OTU-specific posterior means and 95% credible intervals under PGL and PIND in the same tree order. The dashed vertical line marks zero. Related OTUs had broadly consistent effect directions under the two models, whereas PGL yielded greater posterior sign certainty across this local neighborhood.

Table 1

Design of the simulation study. The main experiment contains 12 settings from the full factorial combination of phylogenetic alignment, marginal presence probability, and signal variance. PGL and PIND are fitted to the same simulated data in every replicate.

Component	Main experiment	Sample-size robustness
Sample size, N	50	100
Number of taxa, K	200	200
Target / nuisance covariates	1 / 5	1 / 5
Sequencing depth	50,000	50,000
Active taxa	60 (30%)	60 (30%)
Phylogenetic alignment, A	0, 0.5, 1	0, 1
Presence probability, P	0.25, 0.50	0.50
Signal variance, V	0.50, 1.00	1.00
Methods	PGL, PIND, DESeq2	PGL, PIND
Target replicates per setting	100	100
Primary summaries	Power, FPR, FDR, RMSE	Power, FPR, FDR, RMSE